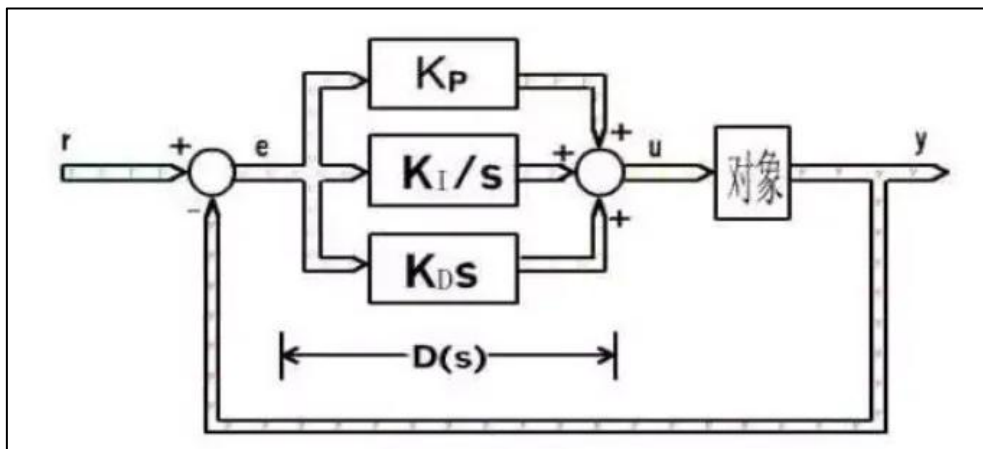


Positional PID Principle

An encoder is a sensor designed to measure the position of a motor. The principle involves detecting changes in the magnetic field of the sensor as the motor rotates, thereby determining the motor's position. During motor rotation, the encoder generates a binary code that reflects the current position of the motor.



PID control is a widely employed closed-loop control algorithm utilized for regulating the speed or position of a motor. The PID controller comprises three components:

Proportional (P): The proportional part adjusts the output in response to the disparity between the motor position and the target position.

Derivative (D): The derivative part modifies the output based on the rate of change in position.

Integral (I): The integral part fine-tunes the output according to the accumulated differences in position.

In motor PID control, the encoder measures the actual motor position. The PID controller calculates based on the disparity between the target position and the actual position, generating a control signal to govern the motor's movement.

The PID controller can adapt the control signal in response to variations in the actual position, enhancing motor control.